

Results of ruthenium irradiation of uveal melanomas: the Dutch experience

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Abstract

Background and purpose: To update our results of the treatment of uveal melanomas using ruthenium applicators in 49 patients treated with graded doses and subsequently in 52 patients with maximal scleral doses of 800 Gy and an effective top dose of at least 160 Gy.

Material and methods: One hundred and one patients were treated with brachytherapy only, in 25 patients it was combined with transpupillary thermotherapy (TTT).

Results: A complete remission was found in 62.6% of the patients and in 31.3% a stable disease with an average follow-up of 74.6 months. Above a topdose of 120 Gy only in one of 95 patients continuous tumour growth after treatment was observed. Useful vision could be preserved in 51.5% of the patients. The initial tumour prominence and top dose strongly correlated with treatment outcome.

Conclusions: Ruthenium application for uveal melanomas with the doses we have used is successful, with a substantial number of patients having their eyes preserved, their tumour controlled and their vision retained. Further improvements can be obtained with ruthenium irradiation with lower maximal scleral doses combined with TTT. © 1999 Elsevier Science Ireland Ltd. All rights reserved.

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1. Introduction

The incidence of uveal melanomas in Western Europe is 0.7 per year per 100 000 persons. In the Netherlands about 100 new cases are diagnosed yearly. In Europe Lommatzsch pioneered the treatment of uveal melanomas in 1964 using ruthenium applicators, introducing these beta-emitting nuclides with directives for treatment [6–9]. As his results were good many centers in Europe started to use these applicators in the late 1970s and early 1980s [1–3,5,10,14–18,20].

In the Netherlands, the treatment of uveal melanomas with irradiation using ruthenium applicators started in 1984, with the University Hospital Leiden as a national referral site. In 1993 we described our results in 49 patients [20]. A dose escalation study was performed with maximal scleral doses from 220 to 950 Gy. The effective dose in the top of the tumour, calculated from the pre-treatment tumour

prominence, including 1 mm of scleral thickness, proved to be an important prognostic factor for tumour control. The best results were obtained with an effective dose in the top of the tumour of at least 160 Gy. The scleral dose seemed to correlate with the probability of complications suggesting that a dose of 800 Gy resulted in an effective top dose for most tumours and an acceptable complication rate.

For treatments as from January 1991 we therefore initiated a protocol aiming at a maximal scleral dose of 800 Gy and an effective top dose of at least 160 Gy. During this period Oosterhuis et al. also conducted a study on the combination of irradiation with the recently introduced transpupillary thermotherapy (TTT) [4,11,12].

2. Material and methods

Between 1984 and 1995 101 patients could be analyzed with a total number of 108 ruthenium applications; seven patients had a second irradiation. Mostly small and medium size melanomas (T1–T2: diameter ≤ 15 mm or height ≤ 5 mm) were irradiated, limited by the size of the applicators

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